

EXPERIMENT NO. 01

TITLE: BASIC FILTER DESIGN



EC39001: Analog Communication Lab

Year: 2026

Department of E&ECE

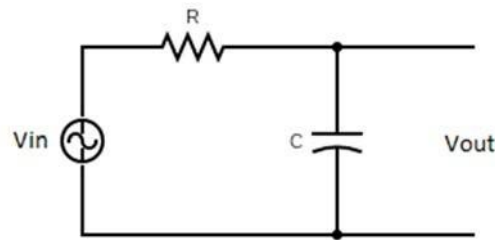
IIT Kharagpur

Note:

1. Before asking for components, show the designed circuit to your allocated TA and get it signed.
2. Assume that Group number = G.

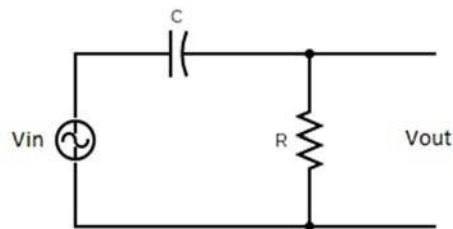
Task 1. Design 1st order passive filter with following corner frequencies:

- a. Low pass filter with cut-off frequency $\{f_H = 4[(G - 1) \bmod 4] + 10\} \text{ KHz}$



Circuit 1: LPF Passive

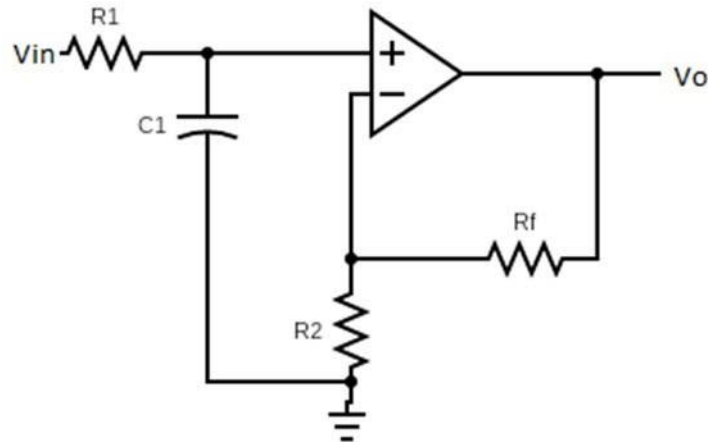
- b. High pass filter with cut-off frequency $\{f_L = 4[(G - 1) \bmod 4] + 4\} \text{ KHz}$



Circuit 2: HPF Passive

- c. Band pass filter with above f_L and f_H .
- d. Find out the 3dB frequency.
- e. Find out the amplitude of signal at $f_L/2$, $2f_H$.

Task 2. Implement 1st order active LPF using Op-Amp with above specifications and a gain of 2. Observe the output on DSO.



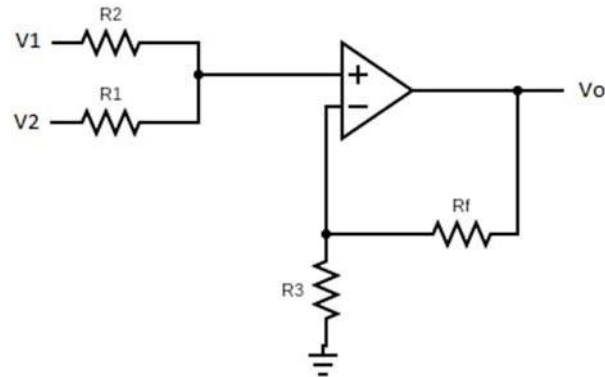
Circuit 3: LPF Active

$$Gain = \left(1 + \frac{R_f}{R_2} \right)$$

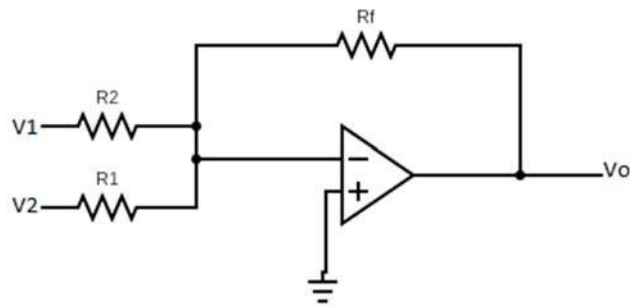
$$f = \frac{1}{2\pi\sqrt{RC}}$$

Task 3. Take two voltage signals $x_1(t)$ and $x_2(t)$ and add them using Op-Amp with following formula:

$$(-1)^G \{x_1(t) + [(G-1) \bmod 4 + 1]x_2(t)\} \text{ Volts}$$



Circuit 4: Adder for even G



Circuit 5: for odd G

